

Title:-

Taking help from your teachers, make a documentary about ISRO the institute mentioned in Chapter 18, using information communication devices.

Introduction:-

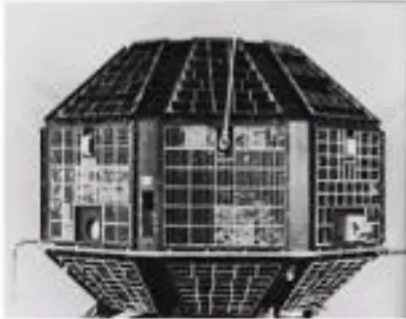


The Indian Space Research Organisation (ISRO) is India's national space agency, responsible for India's journey to space exploration and scientific research. Founded in 1969 under the guidance of Dr. Vikram Sarabhai, ISRO has played a crucial role in making India a leader in space technology and scientific research. ISRO's main goal is to use space technology for the growth and development of the country, benefitting millions through advancements in communication, weather forecasting, disaster management, and agriculture.



Key Milestones in ISRO'S History:-

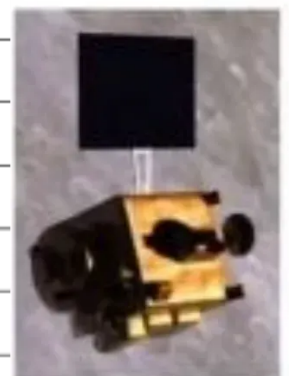
ISRO has achieved significant milestones in its history. Some of the key milestones include:



◆ 1975 Aryabhata: ISRO launched Aryabhata, India's first satellite, named after the ancient Indian mathematician and astronomer. This marked India's entry into space research.



◆ 1980 Rohini: India launched Rohini using its first satellite launch vehicle, SLV-3. This was a major achievement, as it was the first satellite successfully placed in orbit by an Indian-built rocket.



◆ 2008 Chandrayaan-1: This was India's first mission to the Moon. The mission provided valuable data and led to the discovery of water molecules on the Moon's surface.

◆ 2013 Mangalyaan (Mars Orbiter Mission):



2013 Mangalyaan (Mars Orbiter Mission): ISRO's first interplanetary mission to Mars made India the first nation to reach Mars in its first attempt, earning global recognition for ISRO's cost-effective mission.

➤ Structure and Key Centres of ISRO:-

ISRO's headquarters is located in Bengaluru, Karnataka. The organisation is supported by several specialized centers across India, each focusing on different aspects of space technology:

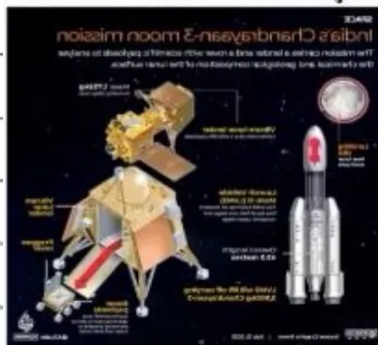


★ Vikram Space (VSSC): Located in Thiruvananthapuram, this center specializes in rocket development and launching vehicles.

★ Satish Dhawan Space Centre (SDSC):



SDSC, Located at Sriharikota, this center manages operations, launch including preparing and launching satellites into space.



★ 2023 Chandrayaan-3: This mission achieving a soft landing near the Moon's south pole, making India the first country to land in this region.

🎯 Space Applications Centre (SAC):



Based in Ahmedabad, SAC works on satellite communications, weather forecast-ing, and remote sensing technologies.

🎯 Liquid Propulsion Systems Centre (LPSC):



LPSC, Located in Valiamala, Kerala, LPSC focuses on rocket propulsion systems and fuels, an essential part of ISRO's launch vehicles.

✓ Satellite Applications and Programs:-

ISRO's satellite programs have numerous applications that benefit millions of people across India. Some of the notable satellite programs are:

- INSAT (Indian National Satellite System): A series of communication satellites that provide TV broadcasting, weather forecasting, and emergency communication.
- GSAT (Geostationary Satellites): These satellites provide internet, mobile communication, and telemedicine services to remote areas, enhancing connectivity and healthcare access.

💡 Impact of ISRO's Missions:-

- ☐ ISRO's missions have had a significant impact on India's growth and development. The benefits include:
- ☐ Communication: ISRO's satellite programs have enabled access television, radio, and internet services, even in remote areas.
- ☐ Weather Forecasting and Disaster Management: Weather satellites help predict weather patterns, cyclones, and manage disaster responses, saving lives and property.
- ☐ Agriculture and Environment: Remote sensing satellites monitor agricultural activities, assess soil quality, and help manage water resources, which is crucial for India's economy.

Conclusion:-

ISRO has played a vital role in placing India among the top space-faring nations of the world. Through innovative and cost-effective missions, ISRO has achieved remarkable milestones, transforming India's capabilities in space science. Its contributions have not only advanced India's technological capabilities but also improved the lives of millions by supporting communication, agriculture, disaster management, and more. With ambitious goals and an inspiring legacy, ISRO continues to inspire and push the boundaries of space exploration.